RELATIVE CYCLES AGREEING OVER GENERIC POINTS ARE EQUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Specialization criterion via DVR-valued morphisms on locally Noetherian schemes (Properties, Lemma properties-lemma-locally-Noetherian-specialization-dvr, canonical tag 054F). cited at proof lines 1100-1104. Supplies g : S' -> S from a DVR with generic point to eta and closed point to s.
- Section on relative fields and base change maps for cycles (relative-cycles chapter, Section section-relative-fields, canonical tag 0H4E). cited at proof lines 1109-1110. Discusses the injectivity of the base change map Z_r(X_s) -> Z_r(X_{0'}).

Source reference. Original lemma 054F: sources/stacks-chow/tags.txt, source lines 6638–6638. Cited at proof lines 1100-1104: supplies the DVR-valued morphism g : S' -> S.

Source reference. Original section 0H4E: sources/stacks-chow/tags.txt, source lines 21349–21349. Cited at proof lines 1109-1110: injectivity of the base change map Z_r(X_s) -> Z_r(X_{0'}).

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S is locally Noetherian and f locally of finite type. Let $r \geq 0$. Let α and β be relative r-cycles on X/S. The following are equivalent*

(1) *$\alpha = \beta$, and*
(2) *$\alpha_\eta = \beta_\eta$ for any generic point $\eta \in S$ of an irreducible component of S.*

Proof. The implication (1) $\Rightarrow$ (2) is immediate. Assume (2). For every $s \in S$ we can find an η as in (2) which specializes to s. By Properties, Lemma 054F we can find a morphism $g : S' \to S$ from the spectrum S' of a discrete valuation ring which maps the generic point η' to η and maps the closed point 0 to s. Then α_s and β_s are elements of $Z_r(X_s)$ which base change to the same element of $Z_r(X_{0'})$, namely $sp_{X_{S'}/S'}(\alpha_{\eta'})$ where $\alpha_{\eta'}$ is the base change of α_η. Since the base change map $Z_r(X_s) \to Z_r(X_{0'})$ is injective as discussed in Section 0H4E we conclude $\alpha_s = \beta_s$. □

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.